

Celiac Disease and the Gluten-Free Diet: Consequences and Recommendations for Improvement

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Key Words

Celiac disease · Gluten-free diet · Micronutrients ·
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Abstract

Background: Celiac disease (CD) is a chronic small intestinal immune-mediated enteropathy precipitated by exposure to dietary gluten in genetically susceptible individuals. CD-related enteropathy leads to multiple nutritional deficiencies involving macro- and micronutrients. Currently, medical nutrition therapy consisting of the gluten-free diet (GFD) is the only accepted treatment for CD. **Key Messages:** The GFD is the cornerstone of treatment for CD. Prior published studies have concluded that maintenance of the GFD results in improvement of the majority of nutritional deficiencies. In the past, counseling for CD focused mainly on the elimination of gluten in the diet. However, the GFD is not without its inadequacies; compliance to the GFD may result in certain deficiencies such as fiber, B vitamins, iron, and trace minerals. Paucity of fortified gluten-free foods may be responsible for certain deficiencies which develop on the GFD. Weight gain and obesity have been added to the list of nutritional consequences while on the GFD and have been partially attributed to hypercaloric content of commercially available gluten-free foods. Follow-up of patients diag-

nosed with CD after starting the GFD has been reported to be irregular and, hence, less than ideal. **Conclusions:** Monitoring of the nutritional status using blood tests and use of appropriate gluten-free supplementation are integral components in the management of CD. The ideal GFD should be nutrient-dense with naturally gluten-free foods, balanced with macro- and micronutrients, reasonably priced, and easily accessible. Rotation of the pseudo-cereals provides a good source of complex carbohydrates, protein, fiber, fatty acids, vitamins and minerals. Fortification/enrichment of commonly consumed gluten-free commercial grain products should be encouraged. Dietitians specializing in CD play a critical role in the education and maintenance of the GFD for patients with CD.

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Introduction

Celiac disease (CD) is defined as a chronic small intestinal immune-mediated enteropathy precipitated by exposure to dietary gluten in genetically predisposed individuals [1]. CD has been reported around the globe with a prevalence of 1–2% of the population in North and South America, North Africa, the Middle East and India [2, 3]. There is a paucity of reliable data regarding the

prevalence in Sub-Saharan Africa and in the Asia-Pacific region [4]. The prevalence of CD has been increasing in the US as noted by a recent study which showed a 5-fold increase since 1975 [5]. Though the reasons for these changes are unclear, they may have to do with the environmental components of CD (changes in the quantity and quality of ingested gluten, infant feeding patterns, the spectrum of intestinal infections, gut microbiota colonization, etc.) [6]. Some of the increase in prevalence can also be attributed to increased awareness and improved diagnostic and epidemiologic tools. The economic burden of undiagnosed CD over 10 years was estimated to be EUR 4 billion in the Mediterranean region alone [7].

Gluten is the alcohol-soluble fraction of wheat protein which is consumed in large quantities (10–20 g per day) as part of the normal food intake in Western countries [8]. It includes a group of prolamines found in the wheat (gliadins and glutenins), rye (secalin), barley (hordein) and other closely related grains [9, 10]. In individuals carrying the haplotype DQ2/DQ8, gluten intake results in activation of a cellular immune response which leads to different degrees of small bowel mucosal injury, ranging from mild intraepithelial lymphocytosis to total villous atrophy [11]. Villous atrophy in the small bowel leads to chronic malabsorption of macro- and micronutrients, which is the hallmark of untreated CD.

Gluten-Free Diet

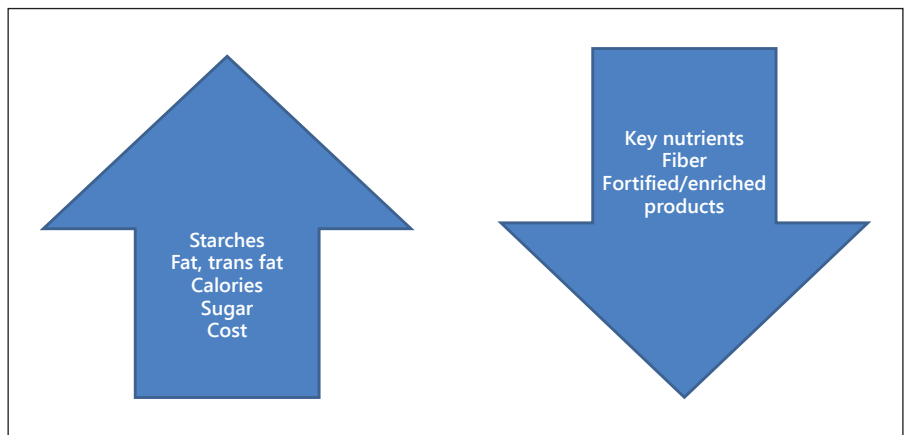
CD is unique amongst chronic immune-mediated disorders in that diet is the only recognized treatment. Currently, initiation and lifelong maintenance of the gluten-free diet (GFD) remains the cornerstone of therapy in CD. The National Institute of Health guidelines on CD emphasize consultation with a skilled registered dietitian (RD) as being a key feature in adhering to the GFD [12]. Learning to follow the GFD often involves major changes in lifelong dietary patterns [13]. Wheat, rye and barley serve as the primary grains in most cereal-based foods in North America and Europe, including breads, cakes, cookies, pastries, pasta, breakfast cereals and many snack foods. In addition, wheat is frequently used as a binder in many soups, sauces and dressings, deli meats, sausages, meat patties and artificial seafood, and as a coating on meat, poultry and fish. Most soy sauce is wheat-based and barley malt is used in the preparation of beer [14].

Adherence to the GFD leads to regression of symptoms, normalization of histological and laboratory findings and reduction in the risk of CD-related complications in most individuals with CD [15]. Zarkadas et al. [16] reported improvement of symptoms in 83% of the respondents with CD on the GFD in a mail-out survey.

It was generally assumed that patients with CD adhering to a strict GFD for years would consume a diet that was nutritionally adequate [17]. However, in the last two decades, multiple studies have shown that patients on the GFD need to be monitored for certain nutritional deficiencies which may complicate the GFD [18–20]. It is well established that the nutritional impact of CD depends on the duration of the untreated disease, extent and location of mucosal injury, and the degree of malabsorption of certain nutrients [21]. In the last 20 years, an additional factor has come into the picture – ‘quality of the GFD’. It is the objective of this paper to highlight the nutritional imbalances of the GFD and suggest measures to overcome them (fig. 1).

Macronutrient Imbalances in the GFD

Often, patients with CD overcome the restriction of the GFD by substituting foods with high fat, sugar and calorie content [22]. Gluten-free products often contain a greater carbohydrate and lipid content than their gluten-containing equivalents. Segura and colleagues reported that the nutritional composition of a range of gluten-free breads had a high glycemic index, with low protein and high fat content. Caponio et al. [23] showed that commercially available gluten-free biscuits were richer in saturated fatty acids compared to the gluten-containing equivalents. Mariani et al. [22] reported in a study involving adolescents with CD that these subjects had a higher fat and protein intake than recommended. In their study, it was proposed that patients following a strict GFD preferred food with the higher amount of protein (meat, eggs, legumes) and snacks with a high content of lipids. The authors also speculated that these dietetic habits in adolescents could cause a disturbance in the growth pattern. Wild et al. [24] in a British study reported that the dietary intake of British female patients with CD contained more energy across all macronutrients as compared to the local population. This was attributed to a higher intake of nonmilk extrinsic sugar present in sugary snacks. The authors cautioned that maintaining available carbohydrate intake using naturally occurring products like beans, pulses and starchy



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Fig. 1. Negative factors affecting quality of gluten-free products [19, 33, 37, 48, 52, 53].

vegetables such as a potato or alternative grains should be done without resorting to processed gluten-free sugary foods. Ohlund et al. [25] reported in a Swedish study involving children with CD that dietary fat intake was relatively high when compared to the Nordic nutrition recommendations (NNR 2004). It was observed that Swedish children with CD had a mean intake of 14% saturated fatty acids and less than 4% polyunsaturated fatty acids [25, 26]. The Nordic Nutrition recommendations suggest saturated fatty acids should contribute less than 10% of the total caloric intake and polyunsaturated fatty acid should contribute at least 5–10% of the total caloric intake [26].

Fiber Intake and the GFD

Chronic constipation is a common and significant symptom in patients with CD [27]. Hence, adequate intake of fiber is important for patients with the above presentation. Dietary fiber is a complex mixture of plant materials and molecules that are resistant to digestion. The beneficial effect of a high-fiber diet in preventing diseases like coronary heart disease, diabetes, and colon cancer has been studied in the past [28]. Some studies have reported that the GFD is associated with a lower intake of dietary fiber. Thompson et al. [19] reported in a 2005 study that less than half of the female survey respondents consumed daily recommended amounts of fiber. Nearly 15% of the US adult daily intake of fiber comes from gluten-containing yeast bread. Unfortunately, the vast majority of gluten-free commercial grain-based products (made from refined gluten-free grains and/or starches)

contain significantly less fiber than their gluten-containing counterparts. Two other studies from the UK also reported similar results. In a British study done by Kinsey et al. [29], a reduced intake of dietary fiber was observed in adult patients with CD with similar results in the general population. Wild et al. [24] reported an inadequate intake of dietary fiber in British individuals consuming a strict GFD.

Micronutrient Imbalances in the GFD

Vitamins

Prior studies have documented deficiencies of both fat- and water-soluble vitamins in patients with newly diagnosed CD [30–32]. Logic dictates that once the GFD is instituted, villous atrophy improves and leads to resolution of malabsorption correcting vitamin deficiencies. However, in the last two decades nutritional studies have indicated that some vitamin deficiencies tend to persist years after institution of the GFD [20, 33]. In a study reported from Australia, inadequate intake of folate and thiamine was documented in patients with CD who had been on the GFD for at least 2 years. This discrepancy was attributed to the fact that (1) very few gluten-free breakfast cereals were fortified with folate as compared to the common practice of fortification in wheat-based cereals, and (2) in the case of thiamine, there was mandatory fortification of thiamine in the wheat flour for regular bread which was not the case for gluten-free equivalents [33]. Hallert et al. [20] in a study conducted in Sweden reported that over 50% of the patients with CD had low vitamin levels (vitamin B₆ and folate), and that these patients had

been adhering to a strict GFD for years. An additional finding of this study was an elevated plasma homocysteine level in adult patients with CD, which could imply an independent increased risk for cardiovascular disease like hypertension and hyperlipidemia. In Sweden, neither gluten-free products nor their gluten-containing counterparts are folate enriched and, hence, it was concluded that the major determinant of the poor vitamin status of patients with CD may be their choice of foods.

This finding again emphasizes the importance of the 'quality' of the GFD in the treatment of patients with CD. The same author conducted a double-blind placebo-controlled multicenter trial where patients with CD on a strict GFD for several years were randomized to a daily vitamin supplementation consisting of pyridoxine, vitamin B₁₂, and folic acid for 6 months. Researchers noted that total homocysteine levels dropped 34% in patients with CD on daily vitamin supplementation and were accompanied by significant improvement in well-being [34].

Martin et al. [35] analyzed the dietary intake of 88 German patients with CD on the GFD and found suboptimal intake of vitamins B₁, B₂, B₆ and folic acid. Similar findings have been published from studies in the US. Thompson [36] in a study published in 1999 observed that only 35 of the 368 gluten-free flours, breads, pastas and cereals were enriched with thiamine, riboflavin and/or niacin. As a result of the findings, a concern was raised about the long-term GFD being deficient in the above nutrients. In a similar study in 2005, Thompson et al. [19] noted that of the 58 gluten-free breads, pastas, and cold cereals, only 3 cold cereals were fortified with folic acid. None of the bread products or pastas was enriched with folic acid.

In addition to inadequate intake of B complex vitamins, reduced intake of vitamin D has been documented in two European studies. Ohlund et al. [25] reported reduced intake of vitamin D in Swedish children on the GFD. Similarly, a British study in adult patients with CD showed reduced intake of vitamin D while adhering to the GFD [29].

Minerals and Trace Elements in the GFD

Iron deficiency is highly prevalent in patients with newly diagnosed CD [30]. Correction of this deficiency is likely to improve some of the related symptoms seen in patients with CD, such as fatigue. Some studies have reported decreased iron intake while following the GFD. Thompson [37] noted that 64 of the 83 gluten-free cereal products had lower amounts of iron as compared to their

gluten-containing counterparts. Only 9 of the 58 gluten-free breads, pastas, and cold cereals were fortified with iron. In a recently published study from Australia, the majority of the female patients with CD on the GFD for at least 2 years were consuming inadequate intakes of iron in relation to recommended daily intake [33]. Reduced calcium intake has also been reported in patients following the GFD. Thompson et al. [19] noted that less than one third of the female survey respondents consumed daily recommended amounts of calcium during the recording period. Similar findings of reduced calcium intake while on the GFD have also been reported in other countries [29, 33]. Reduced milk intake secondary to temporary lactose intolerance in patients with active CD is one of the mechanisms which could explain reduced calcium intake [38].

Other minerals and trace elements which have been documented to be inadequate in intake while on the GFD include magnesium, zinc and selenium [24, 33, 35].

Weight Gain on the GFD

In contrast to the 'classic' celiac presentation of malabsorption and weight loss, overweight and obese patients presenting with CD are becoming increasingly prevalent [39]. A retrospective study published in 2006 reported that nearly 40% of patients with CD were overweight at diagnosis with 13% of them falling in the obese range [39].

Recent studies in the last decade have raised concern that initiation and maintenance of the GFD result in weight gain further compounding the above problem. Valletta et al. [40] reported increased body mass index (BMI) after initiation of the GFD, and the percentage of overweight subjects almost doubled while on the GFD. A proposed theory put forward by the authors was incorrect dietary habits induced by unpalatable, expensive commercial gluten-free products and availability of high-fat commercial gluten-free snacks and biscuits. Kabbani et al. [41] reported significant increase in BMI after the GFD initiation. Nearly 22% of patients who were overweight at diagnosis gained weight while on the GFD.

Enhancing Nutrition of the GFD

Avoiding gluten is the single most important step in the management of CD. Emphasis should be placed on selecting naturally gluten-free foods and avoiding over-

reliance on gluten-free processed foods which tend to be low in fiber, vitamins, and minerals, and higher in sugar, fat, and calories. A naturally healthy GFD is rich in fruits and vegetables, labeled gluten-free whole grains and products made from them, legumes and dried beans, lean protein sources, nuts and seeds, low-fat dairy or nondairy alternatives rich in calcium and vitamin D, and healthy fats, including omega-3 fatty acids. The importance of eating a wide variety of colorful fruits and vegetables for their vitamins and minerals, and phytonutrient and antioxidant compounds cannot be overemphasized.

When initiating the GFD, a protein- and energy-rich diet should be encouraged, particularly if the patient has significant weight loss and deficiencies caused by malabsorption. In most patients, a 6-month period is adequate for malabsorption to improve [42]. Thereafter, a more balanced approach is warranted to avoid increased intake of unhealthy fats which is implicated in obesity and cardiovascular disease.

Weight Gain on the GFD

Weight gain, although a sign of improving absorption, is not always welcome, particularly in patients who are overweight at diagnosis. Weight gain following the diagnosis of CD can be attributed to several factors including the healing of the intestines and changes in diet [43]. Dickey and Kearney [39] reported that 55 of the 67 patients who were overweight at diagnosis gained weight at the end of 2 years on the GFD. The problem of excessive weight gain in patients with CD needs to be addressed by all members of the patient's healthcare team including the RD, gastroenterologist, and primary care physician [44].

Some weight loss tips include replacing at least half of the low fiber, highly refined grains and commercially prepared grain foods eaten daily with gluten-free whole grains and products and eating small, healthy meals and snacks at intervals during the day to avoid strong hunger and cravings. Consulting with an RD to create a healthy eating plan that promotes weight loss and engaging in appropriate regular physical activity are highly recommended [43].

Balanced Food Groups

In patients with CD and the general population, recommendations for food groups and caloric intake are similar and should be individualized as needed. For adults

19 years and older, complex and simple carbohydrates should represent 45–65% of total calories; dietary protein should contribute 10–35% of total calories [45]. Good protein choices include dried beans and legumes, nuts and seeds, fish, lean cuts of poultry, beef, and pork, fat-free or low-fat dairy, and pseudo-grains such as amaranth and quinoa. Total fat intake should represent 20–35% [45], and saturated fatty acid intake should comprise less than 10% of daily total calories, respectively, replaced primarily by monounsaturated and polyunsaturated fatty acids [46]. Trans fats, which are implicated in atherosclerosis, should be consumed as little as possible and be limited to less than 1% of the total caloric intake (5 g/day) [21]. While these acceptable macronutrient distribution ranges take into account intake of essential nutrients and reduction of chronic disease risk, there is wide variability in these proportions among individuals.

Micronutrient and Fiber Intake

Many gluten-free foods are made with refined flour and/or starches which are low in fiber due to processing. It is recommended that patients with CD consume labeled gluten-free whole grains and pseudo-cereals and fortified/enriched products made from them to increase the fiber content of the GFD [47]. These include brown rice, wild rice, buckwheat, quinoa, oats, amaranth, millet, sorghum, and teff. It is recommended to choose 6–11 servings of gluten-free grains daily depending on the caloric content of the patient's diet [19]. At least half of these daily grain servings should come from whole-grain sources. Recommendations for adult fiber intake range from 21 to 38 g per day based on age, gender, and health status [46].

In addition to fiber, the gluten-free grains and pseudo-cereals also provide key vitamins and minerals. In a study by Lee et al. [48], three servings of whole/enriched cereals and pseudo-cereals (oats, brown rice, and quinoa) positively impacted the nutrient profile (thiamin, riboflavin, niacin, folate, iron and fiber) of the grain portion of the diet and were more cost-effective than processed gluten-free grain products. In women of childbearing age, special effort should be made to include nongrain sources of iron and B vitamins during pregnancy and lactation.

The importance of calcium and vitamin D cannot be overemphasized, especially considering the prevalence of metabolic bone disease in the celiac population. It is recommended to select three servings/day of low-fat or nonfat dairy or calcium- and vitamin D-fortified non-

of the GFD [27, 51]. Indisputably, the GFD is the cornerstone of treatment for CD. Keeping these nutritional concerns in mind, a patient with CD can enjoy a healthy, well-rounded diet that improves and maximizes overall health and well-being.

Disclosure Statement

Thimmaiah G. Theethira has no conflict of interest to declare. Melinda Dennis, MS, RD, LDN, is co-author of *Real Life with Celiac Disease: Troubleshooting and Thriving Gluten-Free* published by AGA Press (2010).

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